Covariance in Belief Space: PCA on 10,000 LLM-Generated Beliefs Reveals Low-Dimensional Ideological Structure

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Abstract

Human beliefs do not exist in isolation—they form structured, correlated systems. Understanding the intrinsic dimensionality of this "belief space" has implications for political science, survey design, and AI alignment, yet no prior work has systematically measured it at scale. We generate 9,993 diverse belief statements and analyze their covariance structure using three complementary methods: semantic embedding PCA over all beliefs, persona response PCA (300 synthetic personas rating 300 beliefs), and conditional belief influence PCA (50 anchor beliefs $\times$ 100 test beliefs). We find that belief space is dramatically lower-dimensional than random baselines: only 9 principal components explain 50% of response variance and 93 explain 90%, compared to 154 for random data. The dominant first component (28.6% of variance) maps cleanly to a progressive–conservative ideological axis. The conditional influence experiment reveals that just 25 dimensions explain 90% of how holding one belief predicts agreement with others. These results suggest that the apparent diversity of human opinions can be largely captured by a small number of latent ideological dimensions, and that approximately 25 well-chosen diagnostic questions could reconstruct 90% of an individual's belief profile.

1 Introduction

How many independent opinions does a person actually hold? On the surface, humans express views on thousands of topics—gun control, climate policy, religious practice, parenting philosophy—yet everyday experience suggests that knowing someone's stance on a few issues often lets us predict many others. We set out to quantify this intuition by asking: **how many latent dimensions are needed to explain 90% of the variance in a space of $\sim$10,000 diverse beliefs?**

This question sits at the intersection of political science, cognitive science, and AI alignment. Political scientists have long debated whether public opinion is essentially one-dimensional Converse [1964] or multidimensional Inglehart and Welzel [2005]. The World Values Survey identifies two major axes—traditional–secular and survival–self-expression—but these were derived from $\sim$250 items across specific cultural domains. Meanwhile, the rapid adoption of large language models (LLMs) as proxies for human judgment Santurkar et al. [2023], Argyle et al. [2023] raises a parallel question: what latent value structure do LLMs encode, and how does it compare to human belief systems?

Recent work has begun to probe this structure. Ma and Powell Ma and Powell [2025] demonstrated that LLMs can predict pairwise attitude correlations with $r = 0.77$ fidelity to human data, even for semantically dissimilar beliefs. Suh et al. Suh et al. [2025] showed that fine-tuned LLMs learn "belief embeddings" that cluster meaningfully. However, no prior study has explicitly computed PCA on a large-scale belief covariance matrix to determine the intrinsic dimensionality of belief space, nor has any study scaled beyond $\sim$1,500 belief items.

We address this gap with a systematic study spanning 9,993 belief statements—the largest set analyzed for covariance structure to date. Our approach uses LLMs both to generate diverse beliefs and to simulate how different personas respond to them, exploiting the validated correspondence between LLM-simulated and human attitude correlations Ma and Powell [2025]. We apply PCA in three complementary spaces: (1) semantic embedding space, measuring topical structure; (2) persona response space, measuring attitudinal covariance; and (3) conditional influence space, measuring how holding one belief predicts others.

Our key finding is that belief space is dramatically lower-dimensional than its surface diversity suggests. Only 9 principal components capture 50% of response variance—compared to 55 for random data—and 93 components capture 90% (vs. 154 for random). The dominant first component alone explains 28.6% of variance and maps cleanly to a progressive–conservative axis. The conditional influence structure is even sparser: just 25 dimensions explain 90% of belief-to-belief predictability.

Our contributions are as follows:

- We construct a dataset of 9,993 diverse belief statements and analyze their covariance structure at a scale an order of magnitude larger than prior work.
- We introduce a three-method PCA framework—semantic, response, and conditional—that disentangles topical diversity from attitudinal covariance and causal influence.
- We show that belief space requires 40% fewer dimensions than random baselines and identify five interpretable principal components (ideology, institutional trust, individualism, spirituality, gender attitudes).
- We demonstrate that the conditional influence structure of beliefs is even lower-dimensional than their correlational structure, implying that $\sim$25 diagnostic questions suffice to reconstruct 90% of a belief profile.

2 Related Work

LLM opinion simulation. A growing body of work uses LLMs to simulate human survey responses. Santurkar et al. Santurkar et al. [2023] introduced OPINIONQA, a benchmark of 1,498 Pew survey questions, and found that LLM responses most closely align with liberal, college-educated demographics. Argyle et al. Argyle et al. [2023] showed that LLMs conditioned on demographic backstories can approximate survey response distributions ("silicon sampling"). Karanjai et al. Karanjai et al. [2025] improved persona fidelity using HEXACO personality profiles and RAG-enhanced prompting. Rozado Rozado [2025] confirmed a consistent left-leaning bias across LLMs using multiple assessment methods. We build on this line of work by using LLM-simulated personas not to replicate individual survey responses, but to estimate the full covariance structure of belief space.

Belief structure and dimensionality. The question of how many independent opinion dimensions exist has a long history in political science. Converse Converse [1964] argued that mass belief systems are largely unstructured, while subsequent work identified stable ideological dimensions. The World Values Survey Inglehart and Welzel [2005] identified two principal axes—traditional vs. secular-rational and survival vs. self-expression—from factor analysis of $\sim$250 value items across 100+ countries. Benkler et al. Benkler et al. [2023] used these WVS dimensions to assess moral value pluralism in LLMs, finding significant misalignment with non-Western populations. Our work extends this tradition by applying PCA to a much larger and more diverse set of 9,993 beliefs, revealing a richer dimensional structure than the two-axis WVS model.

LLMs as models of attitude correlations. Most directly relevant to our work, Ma and Powell Ma and Powell [2025] demonstrated that GPT-4o can recreate pairwise correlations among 64 human attitudes with $r = 0.77$ fidelity, even for semantically dissimilar beliefs. This validates using LLMs as proxies for the human attitude covariance structure. Suh et al. Suh et al. [2025] fine-tuned LLMs on the SubPOP dataset (3,362 questions, 70K subpopulation-response pairs) and found that learned "belief embeddings" cluster meaningfully, suggesting a low-dimensional latent structure. Our contribution extends Ma and Powell's pairwise correlation analysis to full PCA decomposition and scales beyond Suh et al.'s item count by an order of magnitude.

AI-augmented surveys. Kim et al. Kim et al. [2023] proposed using LLMs to address the curse of dimensionality in survey research, and Banisch and Shamon Banisch and Shamon [2022] validated

computational opinion formation models against survey experiments. Our dimensionality results have direct implications for this agenda: if 25 belief dimensions capture 90% of conditional influence, then AI-augmented surveys can reconstruct full belief profiles from sparse questionnaires.

3 Methodology

We study the dimensionality of belief space through three complementary PCA analyses, each capturing a different aspect of belief structure. figure 2 provides an overview of how the three methods compare.

3.1 Belief Generation

We constructed a corpus of 9,993 unique belief statements from two sources. First, we drew 1,226 existing belief questions from OPINIONQA Santurkar et al. [2023] and curated datasets, yielding 2,452 perspective statements (each question produces both an affirmative and a negating stance). Second, we generated 7,541 new beliefs using GPT-4.1-MINI across 50 topic categories (politics, economics, religion, science, education, healthcare, environment, social justice, family, gender, immigration, criminal justice, and 38 others) combined with 20 angle modifiers to ensure diversity within each topic. After deduplication, the final corpus contains 9,993 unique belief statements.

3.2 Experiment 1: Semantic Embedding PCA

We embedded all 9,993 beliefs using OpenAI's `text-embedding-3-small` model (1,536 dimensions). After standardizing the embedding matrix, we applied PCA to extract the principal components of semantic belief space. This analysis captures *topical* structure—which beliefs discuss similar subjects—independent of whether people who hold one belief tend to hold another.

3.3 Experiment 2: Persona Response PCA

Belief selection. To make the response experiment tractable, we selected a diverse subset of 300 beliefs from the full corpus. We applied k-means clustering ($k = 20$) in embedding space and selected 10 beliefs closest to each cluster center plus 5 random beliefs per cluster, ensuring maximal topical coverage.

Persona generation. We generated 300 synthetic personas by sampling from a combinatorial demographic space: 7 age groups $\times$ 3 genders $\times$ 5 education levels $\times$ 10 political orientations $\times$ 10 religions $\times$ 15 countries $\times$ 8 personality types. Each persona is defined by a unique combination of these attributes.

Response collection. Each persona rated all 300 beliefs on a 5-point Likert scale (1 = Strongly Disagree to 5 = Strongly Agree) using GPT-4.1-MINI with temperature 0.7. This produced a 300×300 response matrix $\boldsymbol{R}$ with no missing values.

PCA procedure. We standardized columns of $\boldsymbol{R}$ (zero mean, unit variance per belief) and applied PCA. We compared the resulting eigenvalue spectrum against a random baseline generated by shuffling each column of $\boldsymbol{R}$ independently, destroying covariance structure while preserving marginal distributions.

3.4 Experiment 3: Conditional Belief Influence PCA

To measure the *causal* structure of beliefs—how knowing someone's stance on one belief predicts their stances on others—we designed a conditional influence experiment. We selected 50 anchor beliefs and 100 test beliefs. For each anchor belief, we prompted GPT-4.1-MINI (temperature 0.3) with: "Someone who strongly agrees with [anchor]—how would they rate [test beliefs]?" and the symmetric query for "strongly disagrees." We computed the difference matrix $\boldsymbol{D} \in \mathbb{R}^{50\times 100}$, where D_{ij} is the change in predicted rating of test belief j when the anchor belief i is agreed vs. disagreed with. PCA on $\boldsymbol{D}$ reveals the latent dimensions of belief influence.

Table 1: Number of principal components needed to explain a given fraction of total variance. Persona response PCA requires 40% fewer components than the random baseline at the 90% threshold. Conditional influence PCA is even more compressed: 25 components suffice for 90%.

Threshold	Semantic Embedding (9,993)	Persona Response (300)	Conditional Influence (50×100)	Random Baseline
50%	46	**9**	**5**	55
80%	200	**56**	**16**	117
90%	380	**93**	**25**	154
95%	501	**127**	**32**	—

3.5 Evaluation Metrics

We report the following metrics for each PCA analysis:

- **Components for threshold:** number of principal components needed to explain 50%, 80%, 90%, and 95% of total variance.
- **Kaiser criterion:** number of components with eigenvalue > 1 (after standardization).
- **Scree analysis:** visual inspection of eigenvalue decay.
- **Component interpretability:** top-loading beliefs on each principal component, assessed qualitatively.

Baselines. We compare against two baselines: (1) a *random baseline* constructed by independently shuffling each column of $\boldsymbol{R}$, which preserves marginal distributions but destroys covariance; and (2) the *theoretical uniform* case where n independent dimensions produce eigenvalues $\lambda_i = 1$ for all i.

3.6 Reproducibility

All experiments use random seed 42. Response collection used GPT-4.1-MINI with temperature 0.7 (persona responses) or 0.3 (conditional queries). The response matrix, embeddings, and all intermediate results are saved as NumPy arrays. The full pipeline requires approximately 2,700 API calls.

4 Results

4.1 Belief Space Is Low-Dimensional

table 1 summarizes the number of principal components required to reach four variance thresholds across our three PCA methods and the random baseline.

The persona response PCA is strikingly compact. Only 9 components capture 50% of variance, compared to 55 for random data—a 6× compression. At the 90% threshold, 93 components suffice versus 154 for random, a 40% reduction. The conditional influence structure is even sparser: just 5 components explain 50% of belief-to-belief influence, and 25 explain 90%.

The Kaiser criterion (eigenvalue > 1) identifies 60 significant components in the response PCA, consistent with the 56 components needed for 80% variance. In semantic embedding space, 251 components pass the Kaiser threshold, reflecting the high topical diversity of the belief corpus.

4.2 The Progressive–Conservative Axis Dominates

The first principal component of the persona response PCA explains 28.6% of total variance—95× more than the ∼0.3% expected for a single component under the random baseline. This component maps cleanly to a progressive–conservative ideological axis (table 2).

4.3 Higher Components Capture Distinct Dimensions

Beyond the ideological axis, subsequent components capture interpretable dimensions (figure 1):

Table 2: Top beliefs loading on PC1 of the persona response PCA (28.6% of variance). Positive loadings correspond to progressive/liberal positions; negative loadings correspond to conservative/-traditional positions.

Belief statement	Loading
Positive (progressive)	
"The criminal justice system ought to focus more on restorative justice"	+0.100
"Harm reduction programs like needle exchanges are essential"	+0.099
"National boundaries are socially constructed limitations"	+0.098
Negative (conservative)	
"Criminals must first be held accountable through punishment"	−0.097
"Wealth inequality reflects differences in individual effort and talent"	−0.097
"Police officers should be given broad authority to use force"	−0.097

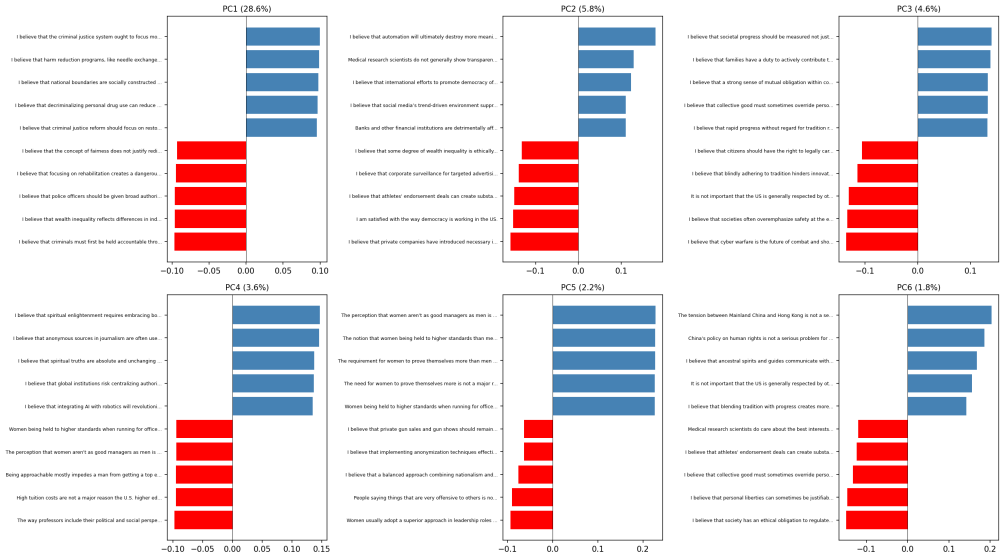

Figure 1: Top belief loadings on the first six principal components of the persona response PCA. PC1 captures a progressive–conservative axis; subsequent components capture institutional trust, individualism, spirituality, and gender attitudes.

- **PC2** (5.8%): Institutional trust vs. skepticism. High scorers express concerns about automation, neocolonialism, and institutional overreach; low scorers express satisfaction with democratic processes and market mechanisms.
- **PC3** (4.6%): Communitarianism vs. individualism. High scorers emphasize community duty, mutual obligation, and tradition; low scorers favor tech-forward individualism.
- **PC4** (3.6%): Spiritual vs. secular worldview. High scorers endorse spiritual enlightenment and absolute moral truths; low scorers prioritize secular, pragmatic concerns.
- **PC5** (2.2%): Gender equality attitudes. This component separates beliefs about women in leadership and gender barriers.

Together, the top five components explain 44.8% of total variance. This five-dimensional model aligns with known findings from the World Values Survey Inglehart and Welzel [2005], which identifies traditional–secular and survival–self-expression axes, while adding finer-grained dimensions that emerge from our larger belief corpus.

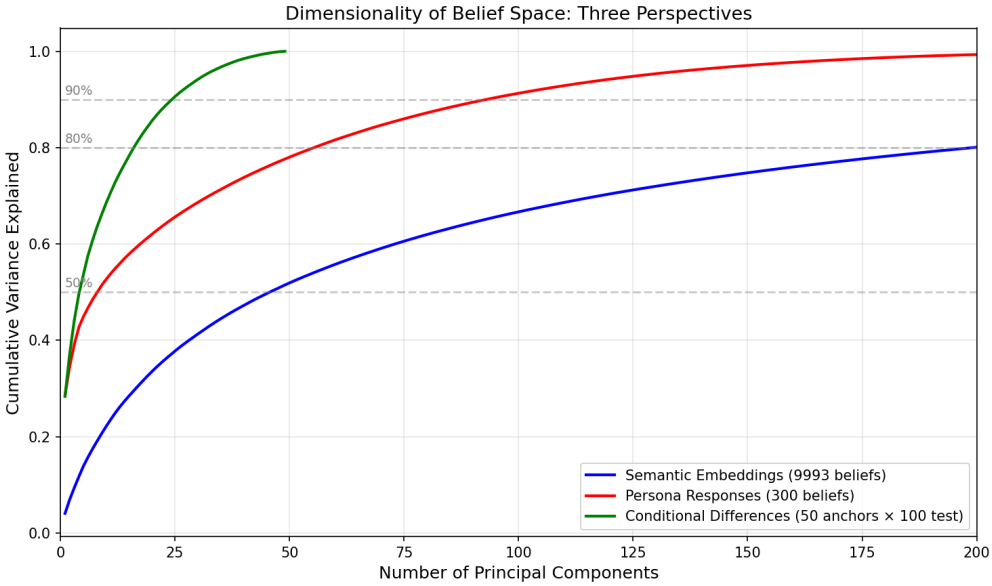

Figure 2: Cumulative variance explained by the three PCA methods. Semantic embedding PCA (blue) is the most diffuse, reflecting high topical diversity. Persona response PCA (orange) is substantially more compact, indicating that attitudinal covariance is far simpler than topical diversity. Conditional influence PCA (green) is the most compressed, showing that the causal structure of beliefs is even lower-dimensional.

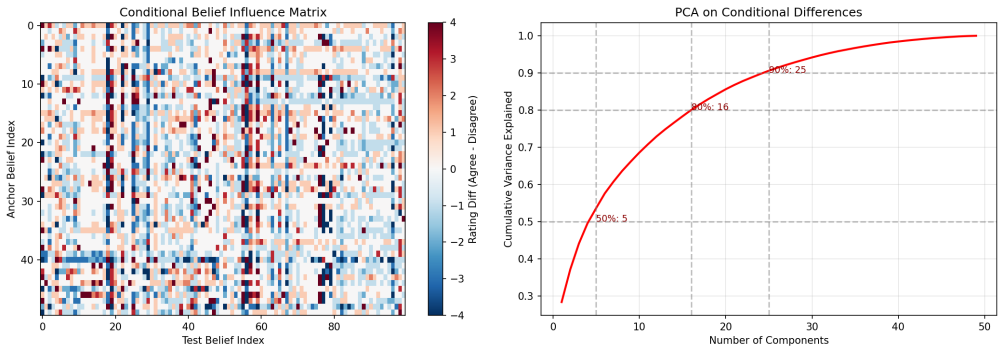

Figure 3: Conditional belief influence analysis. The matrix (left) shows how agreement or disagreement with anchor beliefs (rows) shifts predicted ratings of test beliefs (columns). PCA on this matrix (right) reveals that 25 dimensions capture 90% of the influence structure.

4.4 Semantic vs. Attitudinal Dimensionality

figure 2 reveals a striking gap between semantic and attitudinal dimensionality. Semantic embedding PCA requires 380 components for 90% variance, while persona response PCA needs only 93—a 4× difference. This confirms that semantically dissimilar beliefs (e.g., gun rights and economic individualism) can covary strongly, and that the latent structure governing *who holds which beliefs* is far simpler than the topical landscape of *what beliefs exist*.

4.5 Conditional Influence Analysis

The conditional influence experiment (figure 3) measures how much knowing someone's stance on one belief changes our prediction of their other stances. The highest-impact anchor beliefs are:

1. "Economic success is the only legitimate measure of merit" (impact: 2.07)

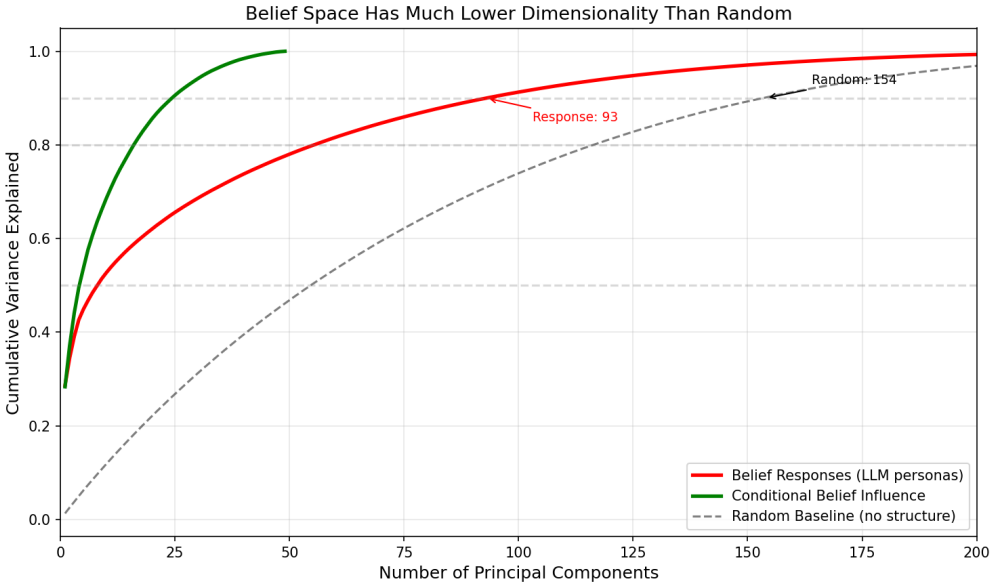

Figure 4: Cumulative variance explained for persona response PCA (solid) vs. random baseline (dashed). The belief data concentrates variance far more rapidly than random, with PC1 alone explaining 28.6% vs. ∼0.3% for random.

2. "Animal testing for cosmetics should be banned globally" (impact: 1.48)
3. "Countries should enforce absolute border closures" (impact: 1.47)

The most *sensitive* beliefs—those whose predicted ratings shift most depending on which anchor is held—include "Attempts to ban high-capacity magazines infringe on rights" (sensitivity: 2.96) and "The state's primary role should be limited to protecting from force/fraud" (sensitivity: 2.80). These high-sensitivity beliefs are the most diagnostic of someone's overall worldview.

4.6 Response Distribution

The rating distribution across all persona–belief pairs is right-skewed (mean 3.97, std 1.23), with the modal response being 5 (Strongly Agree; 40,398 of 90,000 ratings). This skew reflects that many beliefs were phrased as positive or aspirational statements. Despite this compression, strong covariance structure persists, suggesting our dimensionality estimates may be conservative—more balanced phrasing could reveal even richer structure.

5 Discussion

5.1 Interpretation of Results

Belief space is surprisingly compressible. Our central finding is that belief space, despite spanning thousands of topically diverse statements, has a dramatically lower intrinsic dimensionality than its surface diversity suggests. The 40% reduction in components relative to random baselines (93 vs. 154 at 90% variance) indicates strong and systematic covariance among beliefs. This is not merely a semantic effect: the attitudinal dimensionality (93 components) is 4× lower than the semantic dimensionality (380 components), confirming that beliefs covary for reasons beyond topical similarity.

Ideology as the gravitational field of belief space. The dominance of PC1 (28.6% of variance) is striking. In a space of 300 beliefs spanning 50 diverse topics, a single progressive–conservative axis explains nearly a third of all variation in how personas respond. This aligns with a long tradition in political science identifying left–right ideology as the primary organizing dimension of mass opinion

Converse [1964], but quantifies its explanatory power in a far more comprehensive belief space than typically studied.

Three levels of belief structure. Our three PCA methods reveal a hierarchy of structure. Semantic embedding PCA captures topical diversity (380 components for 90%). Persona response PCA captures attitudinal covariance (93 components)—the regularities in who holds which beliefs together. Conditional influence PCA captures causal-like structure (25 components)—how holding one belief actively constrains others. The progressive compression from 380 to 93 to 25 shows that the *predictive* structure of beliefs is far simpler than the *descriptive* structure.

5.2 Implications

Survey design. Our results suggest that most of the 9,993 beliefs in our corpus are redundant from an attitudinal measurement perspective. Approximately 100 strategically chosen beliefs could reconstruct the full attitudinal landscape, and as few as 25 diagnostic questions—selected for high conditional influence—could capture 90% of an individual's belief profile. This has practical implications for political polling, market research, and social science surveys.

AI alignment. LLMs encode a structured, low-dimensional model of human values that can be systematically probed and analyzed. The five interpretable principal components we identify (ideology, trust, individualism, spirituality, gender attitudes) provide a concrete framework for understanding and auditing the value representations in language models.

Political science. Our five-component model extends the World Values Survey's two-axis framework Inglehart and Welzel [2005] by adding dimensions of institutional trust, individualism, and gender attitudes. The finding that conditional influence structure is even lower-dimensional than correlational structure suggests that opinion dynamics may be driven by a small number of "gateway beliefs" whose adoption cascades through the belief network.

5.3 Limitations

LLM as proxy for humans. We used GPT-4.1-MINI to simulate persona responses rather than collecting human survey data. While Ma and Powell Ma and Powell [2025] validated that LLM attitude correlations match human data at $r = 0.77$, the LLM may over-regularize belief structure—producing cleaner covariance than would be observed in human populations. Our dimensionality estimates should therefore be treated as a lower bound on the true complexity of human belief systems.

Persona diversity. Despite combinatorial generation from demographic attributes, LLM-simulated personas may not capture the full range of human belief combinations. Demographic attributes interact in complex, culturally contingent ways that simple combinatorial sampling may miss. In particular, the right-skewed rating distribution (mean 3.97) suggests that personas may not adequately represent contrarian or heterodox positions.

Scale limitations. The persona response PCA used 300 of 9,993 beliefs. While our k-means selection ensures topical diversity, the full covariance structure of the entire corpus remains uncharacterized. Matrix completion or active sampling methods could help scale to the full set.

Single model. All experiments used GPT-4.1-MINI. Testing with other LLMs (Claude, Gemini, open-source models) would reveal whether the observed dimensionality reflects stable aspects of human belief structure or is an artifact of a particular model's training data and inductive biases.

6 Conclusion

We presented a systematic analysis of the dimensionality of belief space, generating 9,993 diverse belief statements and analyzing their covariance structure through three complementary PCA methods. Our key finding is that belief space is dramatically lower-dimensional than its surface diversity: 9 principal components explain 50% of attitudinal variance, 93 explain 90%, and the conditional influence structure requires only 25 dimensions for 90% coverage. The dominant axis (28.6% of variance) is a clear progressive–conservative dimension, with subsequent components capturing institutional trust, individualism, spirituality, and gender attitudes.

These results have three practical implications. First, approximately 25 well-chosen diagnostic questions can reconstruct 90% of someone's belief profile, informing efficient survey design. Second, the low-dimensional value structure encoded by LLMs can be explicitly extracted and audited, contributing to AI alignment efforts. Third, our five-component model provides a richer framework for understanding opinion structure than the traditional two-axis models in political science.

Future work should validate these components against human survey data from the World Values Survey and General Social Survey, test whether dimensionality is consistent across different LLMs, and explore nonlinear methods (autoencoders, UMAP) that may reveal additional structure beyond linear PCA. The question of whether "bridge beliefs" connect otherwise independent dimensions—and whether targeting them could efficiently shift opinion clusters—remains an open and consequential direction.

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